MIL-P-25726B(ASG)
10 JUNE 1959

Superseding
MIL-P-25726A(USAF)
17 July 1957

MILITARY SPECIFICATION

PROBE, TOTAL TEMPERATURE, TYPE MA-1

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers the requirements for one type of total temperature probe, designated Type MA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect
MS3108	Connector, Plug, Electric, Angle 90°
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN509	Screw - Machine, Flat Head, 100°, Structural

FSC 6685

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PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of
No. 505	Storage Life - Aeronautical Articles

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The probe furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Materials.-

3.2.1 Fungus-proof materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.2.2 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.2.3 Corrosive fumes.- The materials as installed in the probe and under the service conditions specified in section 4, shall not liberate deleterious fumes.

3.2.4 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.5 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Standard MS33586.

3.2.6 Protective treatment.- When materials are used in the construction of the probe that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and service not specified herein shall be selected in accordance with ANA Bulletin No. 143 except as provided in 3.3.1 and 3.3.2.

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3.3.1 Commercial parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.
(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.2 Standard parts.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Design and construction.-

3.4.1 The design of the probe shall essentially conform to figures 1 through 6. The design details shall in no way constitute a waiver from the performance requirements specified herein.

3.4.2 The probe shall consist of a resistance element housed in radiation shields, lead wire, insulating strut, and electrical connector.

3.4.3 The probe shall be so constructed that repairs and adjustments can be accomplished by personnel at repair bases without the use of special tools.

3.4.4 The design criteria specified herein which covers the aerodynamic and dynamic performance and dimensional interchangeability of the probe are mandatory unless new design criteria are proposed which will provide equal or better performance and interchangeability. Design criteria, other than that referenced above, may be deviated from to facilitate ease of manufacture. All design changes are subject to the approval of the procuring activity.

3.5 Performance.- The probe shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

(a) Temperature: Temperatures ranging from -70° to +350°C.

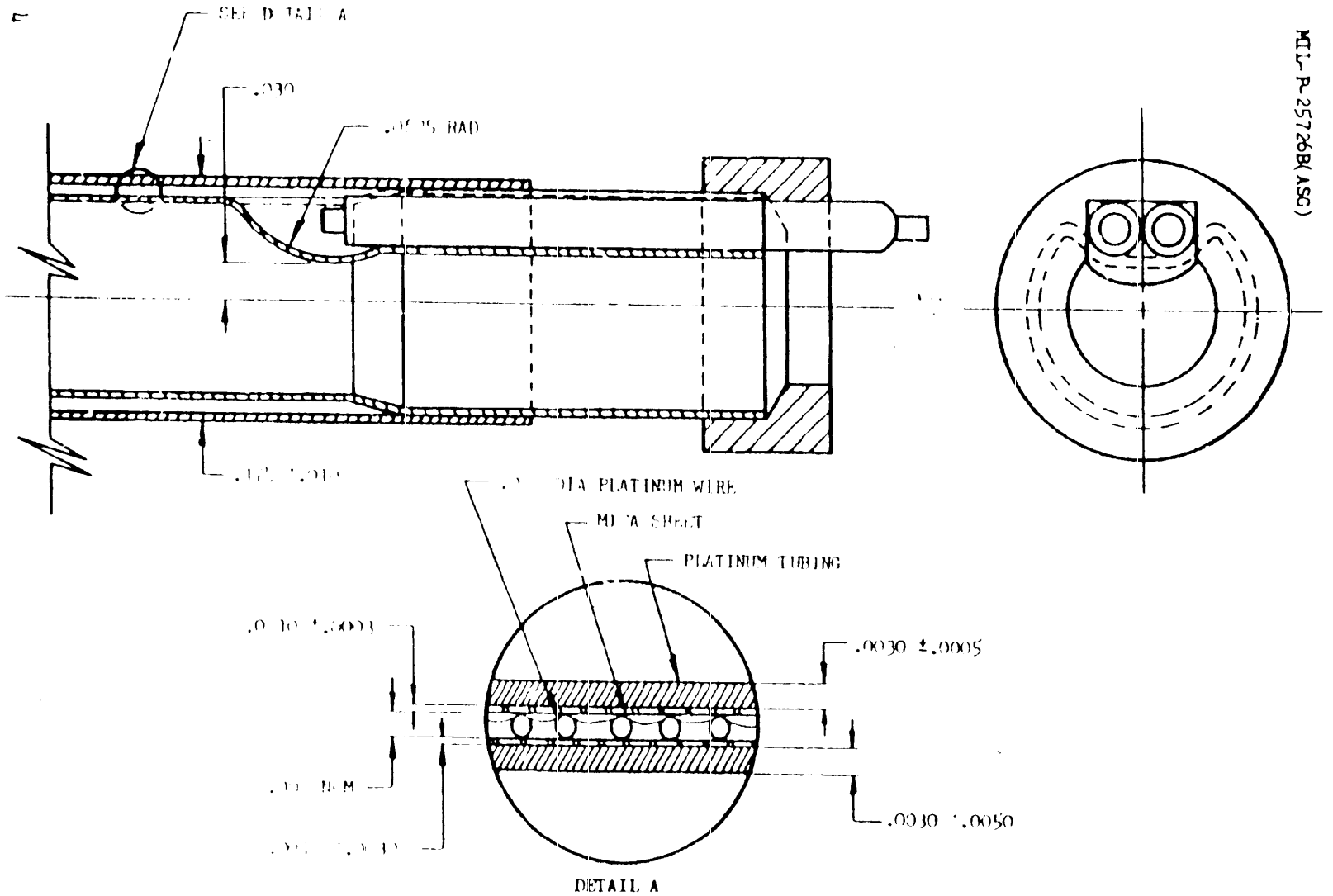
(b) Humidity: Relative humidity up to 100 percent.

(c) Vibration: Vibration at frequencies between 10 and 500 cps at an applied double amplitude of 0.036 inch under the conditions specified for Vibration tests, Procedure XII, of Specification MIL-E-5272.

(d) Thermal shock: Cycled at temperatures of 0° ±5°C and +200° ±5°C alternately for 30-minute periods for a period of 8 cycles.

(e) Temperature-resistance: Measurable resistance over the range of 0°C to the boiling point of water.

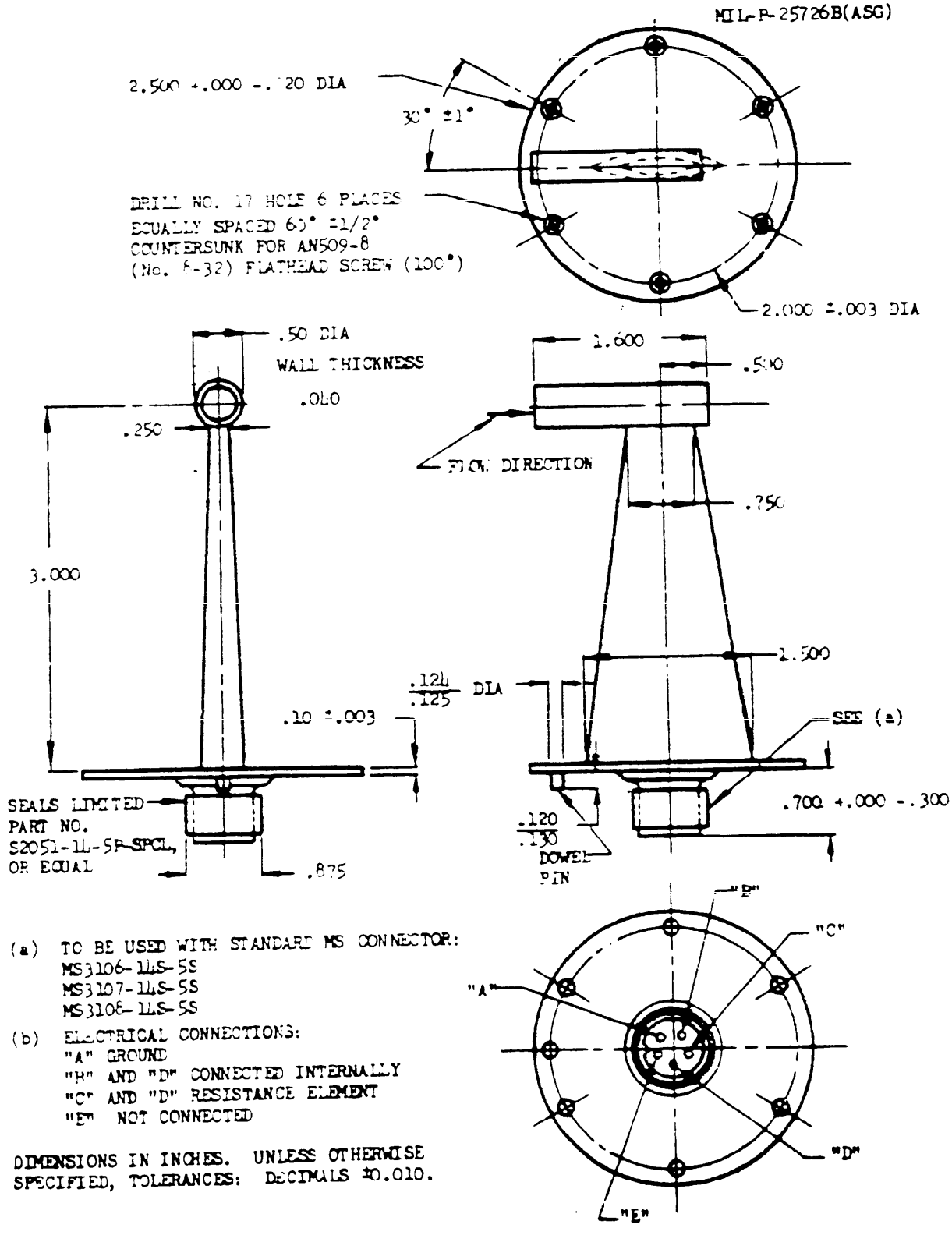
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010.

FIGURE 1. Resistance element.

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(a) TO BE USED WITH STANDARD MS CONNECTOR:
MS3106-11S-5S
MS3107-11S-5S
MS3108-11S-5S

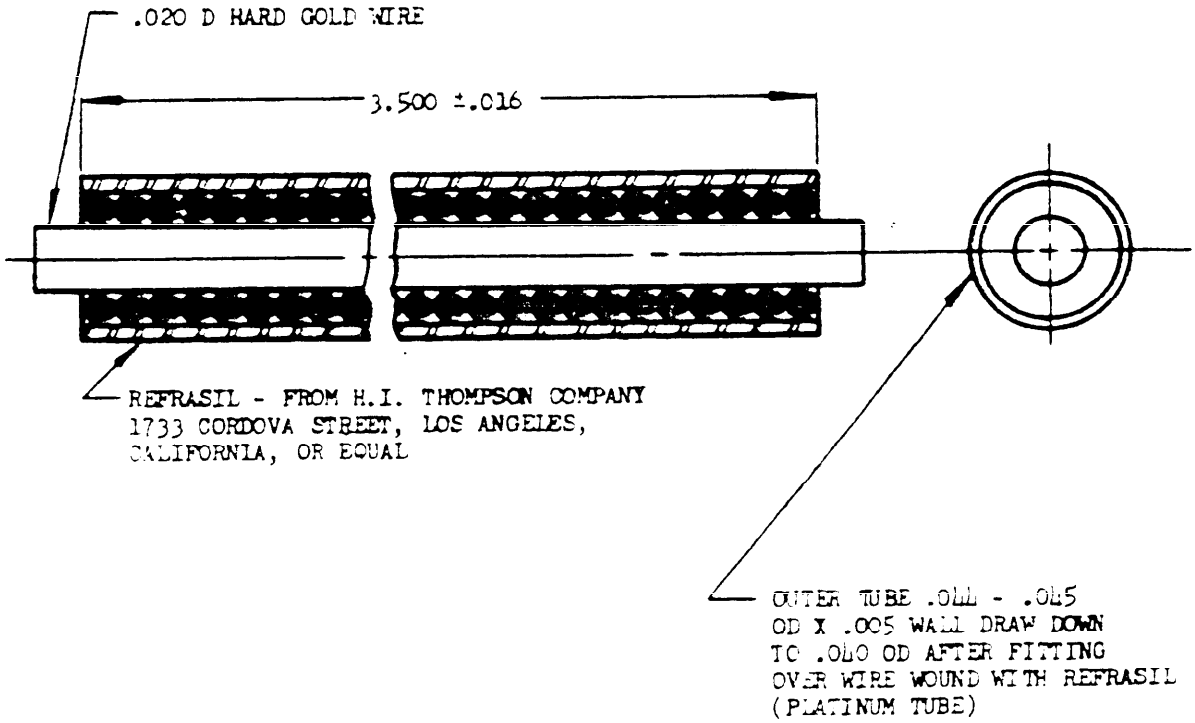
(b) ELECTRICAL CONNECTIONS:
"A" GROUND
"B" AND "D" CONNECTED INTERNALLY
"C" AND "D" RESISTANCE ELEMENT
"E" NOT CONNECTED

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±0.010.

TOTAL TEMP PROBE

FIGURE 2. Mounting layout and electrical connections

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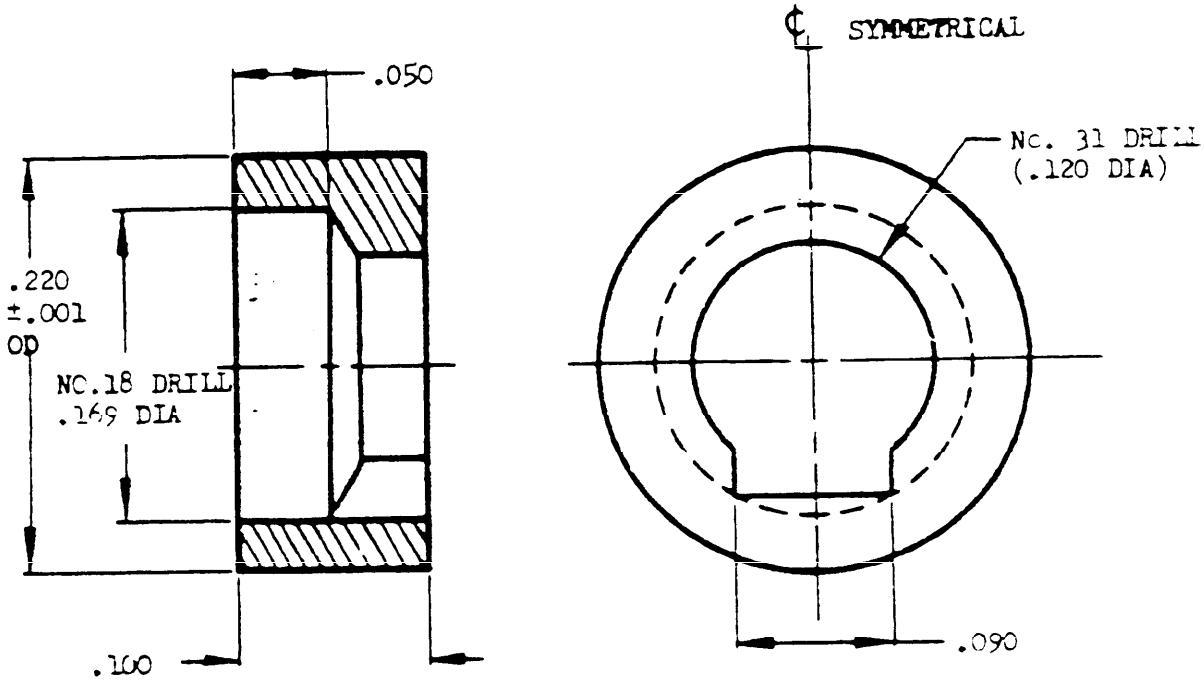


DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±.010.

TOTAL TEMP PROBE

FIGURE 3. Coaxial lead wire

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SCALE: 10 TIMES

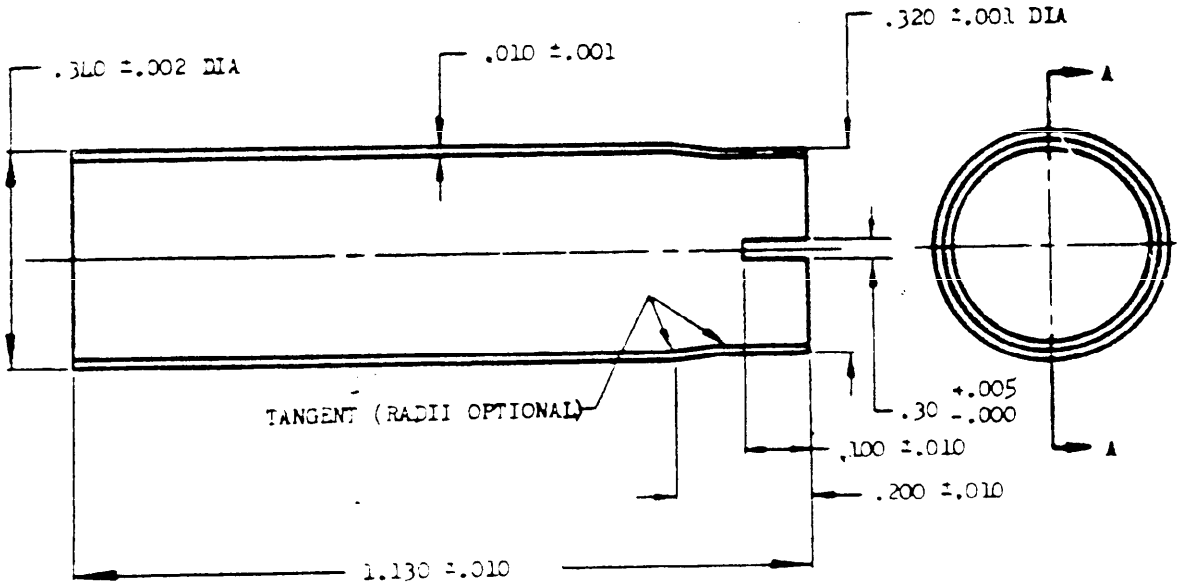
MATERIAL: 90 PERCENT GOLD, 10 PERCENT PLATINUM.

TOTAL TEMP PROBE

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±.010.

FIGURE 4. Sonic throat

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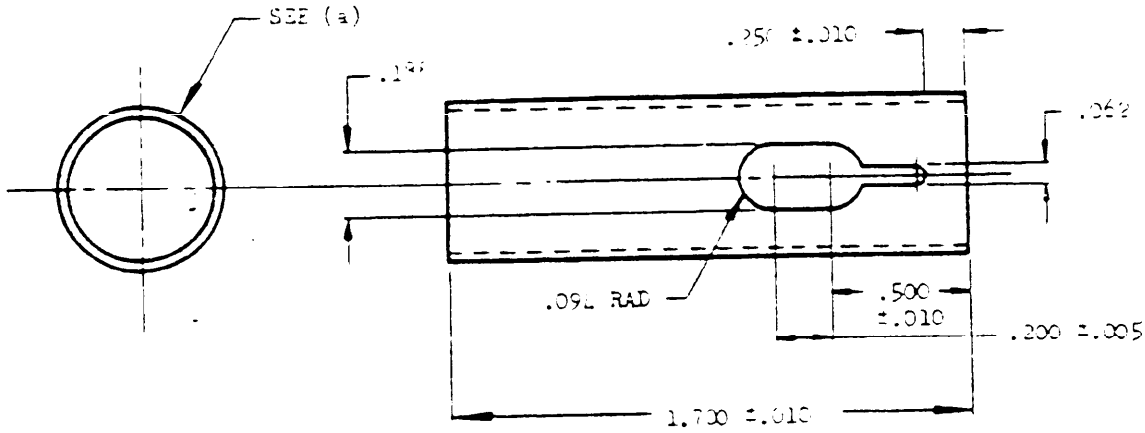
SECTION A-A

TOTAL TEMP PROBE

DIMENSIONS IN INCHES.

FIGURE 5. Inner shield

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(a) STAINLESS STEEL TYPE 304L TUBING
OD .500 +.001 -.000
WALL THICKNESS .020 ±.002

TOTAL TEMP PROBE

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±.016.

FIGURE 6. Outer shield

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(f) Insulation: Satisfactory electrical resistance measurements after being submerged in water for a period of 8 hours.

(g) Thermal response: When subjected to an airflow of Mach 0.3 at sea level, the time constant shall not exceed 1.5 seconds.

(h) Reactance: The resistance shall not vary more than 0.04 ohm at any frequency measurement from 0 to 2,000 cps when tested for inductive or capacitive resistance effect.

(i) Self-heating: Self-heating shall not change the resistance more than 3°C.

(j) Static loading: The probe shall withstand static impact pressures up to Mach 2.0.

3.6 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of Specification MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.7 Resistance element.- The resistance element shall conform to figure 1. Deviations in the design and construction of the resistance winding shall be subject to the approval of the procuring activity.

3.7.1 Electrical characteristics.- The resistance element shall be constructed of pure platinum wire and the resistance versus temperature relationship shall follow the Callendar-Van Dusen equation:

$$R_T/R_0 = 1 + \alpha\left[T - \delta\left(\frac{T}{100}-1\right)\left(\frac{T}{100}\right)^3 - \beta\left(\frac{T}{100}-1\right)\left(\frac{T}{100}\right)^3\right] \qquad (1)$$

Where R_T = resistance at temperature °C, R_0 = resistance at 0°C, α = 0.003925, δ = 1.485, β = 0 (for temperatures above 0°C) and β = 0.10 (for temperatures below 0°C) and T = temperature °C. The resistance element shall be adjusted to 50.00 ohms at 0°C. Deviations from the Callendar-Van Dusen equation over the temperature range of the probe shall be not greater than allowed by the following equation:

$$\Delta T = \pm[0.25 + 0.005\,T] \qquad (2)$$

Where ΔT is allowable deviation in °C. (The negative sign, when T is below 0°, should not be used in this equation.)

3.8 Lead wire.- The design and construction of the lead wire shall normally conform to figure 3. Deviations in the design and construction of the lead wire are acceptable, provided the dimensional requirements of figure 3, the thermal conductivity requirements, and the performance requirements specified herein are met. Deviations in the design and construction of the lead wire shall be subject to the approval of the procuring activity.

3.9 Total temperature range.- The probe shall be designed to operate over a total temperature range of -70°C to +350°C.

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3.10 Operating characteristics.- The probe shall be designed for continuous operation at +250°C and intermittent (1-hour intervals) operation at 350°C. The probe shall be designed to have a minimum operational life of 1,000 hours. The probe will be subjected to temperatures of +350°C for approximately 10 percent of the total operating life. For a temperature of 350°C at the sensing element, the temperature of the 1/2 inch of the strut nearest the connector shall be not lower than 250°C.

3.11 Insulation materials.- Insulation materials utilized in the probe design shall be suitable for the high temperature and shall be essentially non-hygroscopic.

3.12 Weight.- The weight of the probe shall not exceed 200 grams.

3.13 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.14 Workmanship.- The probe shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, welding, brazing, and freedom of parts from burrs and sharp edges.

3.14.1 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances affect interchangeability, operation or performance of the probe, they shall be held or limited accordingly.

3.14.2 Cleaning.- The probe shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material removed after final assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.- The inspection and testing of the probe shall be classified as follows:

(a) Qualification tests (See 4.2.)
(b) Acceptance tests (See 4.3.)

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4.2 Qualification tests.-

4.2.1 Qualification test samples.- The qualification test samples shall consist of three probes. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.2.2 Qualification required.- Prior to actual procurement, the probe which this specification covers shall pass the qualification tests specified herein. If the probe is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.2.3 Tests.- The qualification tests shall consist of all the tests of this specification, as described under 4.5 "Test methods."

4.3 Acceptance tests.- The acceptance tests shall consist of individual tests and sampling plans A and B tests.

4.3.1 Individual tests.- Each probe shall be subjected to the following tests, as described under 4.5 "Test methods":

(a)	Examination of product	(4.5.1)
(b)	Temperature resistance	(4.5.2)
(c)	Insulation	(4.5.3)

4.3.2 Sampling plans A and B tests.-

4.3.2.1 Sampling plan A tests.- One probe selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests, as described under 4.5 "Test methods":

(a)	Individual tests	(4.3.1)
(b)	Temperature calibration	(4.5.4)
(c)	Vibration	(4.5.5)

4.3.2.2 Sampling plan B tests.- Unless otherwise specified, three probes selected at random from the first 15 items of the contract or order shall be subjected to the following tests, as described under 4.5 "Test methods":

(a)	Sampling plan A tests	(4.3.2.1)
(b)	Thermal response	(4.5.6)
(c)	Aerodynamic calibration	(4.5.7)
(d)	Reactance	(4.5.8)
(e)	Self-heating	(4.5.9)
(f)	High-temperature exposure	(4.5.10)
(g)	Low-temperature exposure	(4.5.11)
(h)	Thermal shock	(4.5.12)
(i)	Humidity	(4.5.13)
(j)	Static loading	(4.5.14)

4.3.2.3 Rejection and retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

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4.3.2.3.1 Individual tests may continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.3.3 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions.-

4.4.1 All resistance measurements shall be made by applying 400 cps power. The current flowing through the resistance element shall be of such magnitude that the self-heating effect shall not exceed 0.1°C.

4.4.2 Unless otherwise specified, the probe shall be tested in air except as indicated in 4.5.2 and 4.5.4.

4.5 Test methods.-

4.5.1 Examination of product.- The probe shall be inspected to determine conformance to the requirements specified herein with respect to materials, workmanship, and marking.

4.5.2 Temperature resistance.- The resistance of the probe shall be measured at 0°C and at the boiling point of water. The probe shall meet the temperature resistance calibration equation specified in 3.7.1 with a tolerance of ±(0.25°C +0.005 t°C).

4.5.3 Insulation.- The probe shall be tested by complete submersion in tap water for a period of 8 hours. Upon completion of the 8-hour period, the probe shall be removed from the water and dried for a period of 1 hour. At the end of this time, a potential of 100V dc shall be applied between any electrical receptacle pin contact of the resistance element and any ground part of the probe. The resistance measured shall exceed 10 megohms.

4.5.4 Temperature calibration.- The resistance of the probe shall be measured at 0°C, the boiling point of water, and at a temperature accurate to 0.2°C between 250° and 350°C, and at -40°C. The probe shall meet the temperature-resistance calibration equation specified in 3.7.1 with a tolerance of ±(0.25°C +0.005t°C).

4.5.5 Vibration.- The probe shall be subjected to the Vibration tests, Procedure XII, of Specification MIL-E-5272. This test shall be conducted with the probe at room temperature and at 250° ±25°C. No damage to the probe or change in calibration shall result from this test.

4.5.6 Thermal response.- The time constant shall be determined at an airflow of Mach number 0.3 and sea level pressure. The time constant shall not exceed 1.5 seconds. The thermal time constant is defined as the time required for the

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probe to reach 63 percent of a step function change in temperature and is expressed by the following equation:

$$a-b_1 = (a-b)\, e^{-\tau/T} \qquad (3)$$

Where a = final temperature
b_1 = temperature of probe at time τ
b = temperature of probe at time $\tau = 0$
τ = time in seconds
T = thermal time constant in seconds

4.5.7 Aerodynamic calibration.- The conduction and radiation errors of the probe shall be determined under dynamic conditions. The design of the aerodynamic test apparatus shall be submitted to the contracting officer for approval prior to conducting tests. The probe shall be subjected to the aerodynamic conditions necessary to produce the conditions specified in table I. The errors of the probe in determination of the stagnation temperature shall not exceed the values specified in table I.

TABLE I

Aerodynamic conditions

Stagnation pressure - mm Hg	Stagnation temperature - °C	Allowable error in measured stagnation temperature ±%
1,000	250	0.5
1,000	350	0.5
100	250	0.5
100	350	0.7
30	250	0.7
30	350	1.0

4.5.8 Reactance.- The probe shall be tested for inductive or capacitive reactance effect, or both, on the resistance of the probe. The resistance of the probe shall not vary more than 0.05 ohm at any frequency of measurement from 0 to 2,000 cps.

4.5.9 Self-heating.- The bulb shall be tested in room temperature air for self-heating effect. Sufficient voltage shall be applied to produce a current of 50 milliamperes. This test shall be conducted with an internal flow velocity through the probe at 0.3 Mach. The change in resistance shall not exceed 3°C.

4.5.10 High-temperature exposure.- The probe shall be subjected consecutively to a temperature of +250° ±5°C for a period of 7 hours and a temperature of +350° ±5°C for a period of 1 hour for a total of 155 hours of exposure. After exposure, the resistance of the probe shall be determined at 0°C and +100°C and shall not differ by more than the equivalent of 0.2°C from the resistance values before testing.

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4.5.11 Low-temperature exposure.- The probe shall be subjected to a temperature of -70° ±2°C for a period of 15 hours. No damage to the probe shall result from this test. The resistance of the probe shall be determined at 0°C and +100°C and shall not differ by more than the equivalent of 0.2°C from the resistance values before testing.

4.5.12 Thermal shock.- The probe shall be subjected to a temperature of 0° ±5°C for a period of 30 minutes. The probe shall be removed from this medium and within 10 seconds shall be subjected to a temperature of +200° ±5°C for a period of 30 minutes. The probe shall then be removed from this medium and within 10 seconds resubjected to a temperature of 0°C. This thermal cycle shall be repeated 8 times. The resistance of the probe shall be determined at 0°C and +100°C after test. No damage to the probe or change in calibration shall result from this test by more than the equivalent of 0.2°C.

4.5.13 Humidity.- The probe shall be subjected to the Humidity tests, Procedure I, of Specification MIL-E-5272. There shall be no damage or corrosion of the probe as a result of this test. No change in calibration shall result from this test. At the end of this test, the probe shall be air dried for 1 hour. At the end of this period of time, a potential of 100V shall be applied between any electrical receptacle pin contact of the resistance element and any ground part of the probe. The resistance measured shall be greater than 10 megohms.

4.5.14 Static loading.- The probe shall be statically loaded to simulate the impact pressures produced by Mach 2.0 flow at sea level pressure. No damage to the probe shall result from this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Each probe shall be packaged and packed for shipment in accordance with Specification MIL-P-7936. Preservation shall be in accordance with Method III of Specification MIL-P-116.

5.2 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall include the following:

PROBE, TOTAL TEMPERATURE, TYPE MA-1

5.2.1 Reinspection date.- The reinspection date shall be in accordance with AMA Bulletin No. 405.

6. NOTES

6.1 Intended use.- The Type MA-1 probe covered by this specification is intended for use in aircraft as a means of determining the total temperature developed by adiabatic heating of the air due to the motion of the aircraft through the air.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing. (See 5.2.)

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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